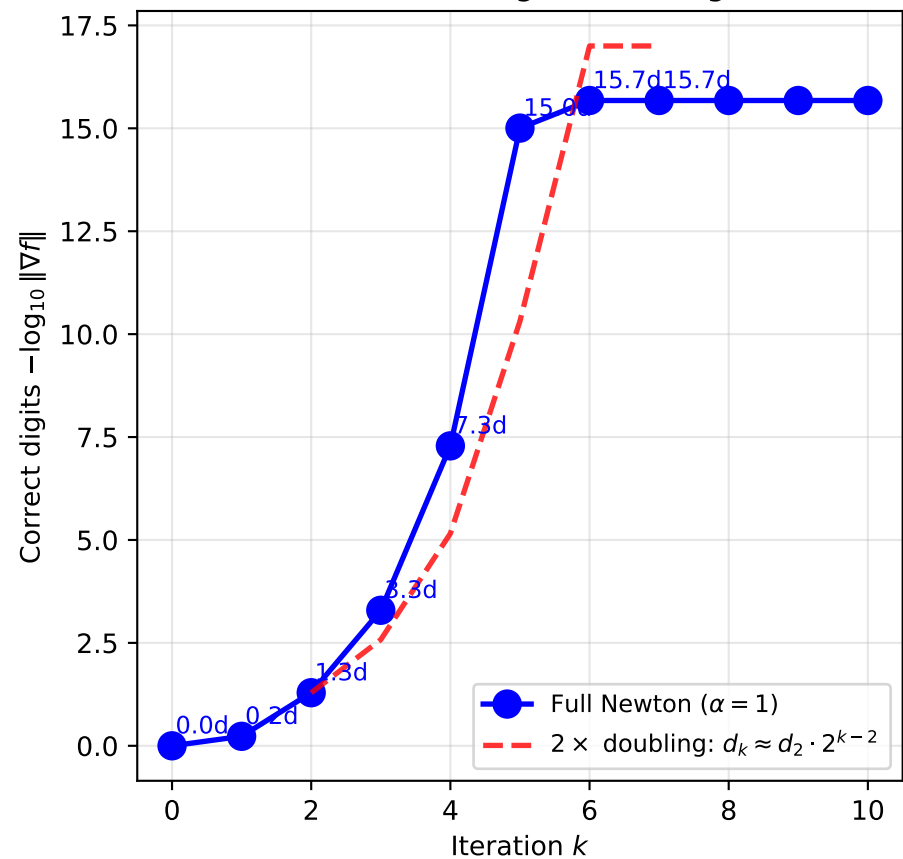
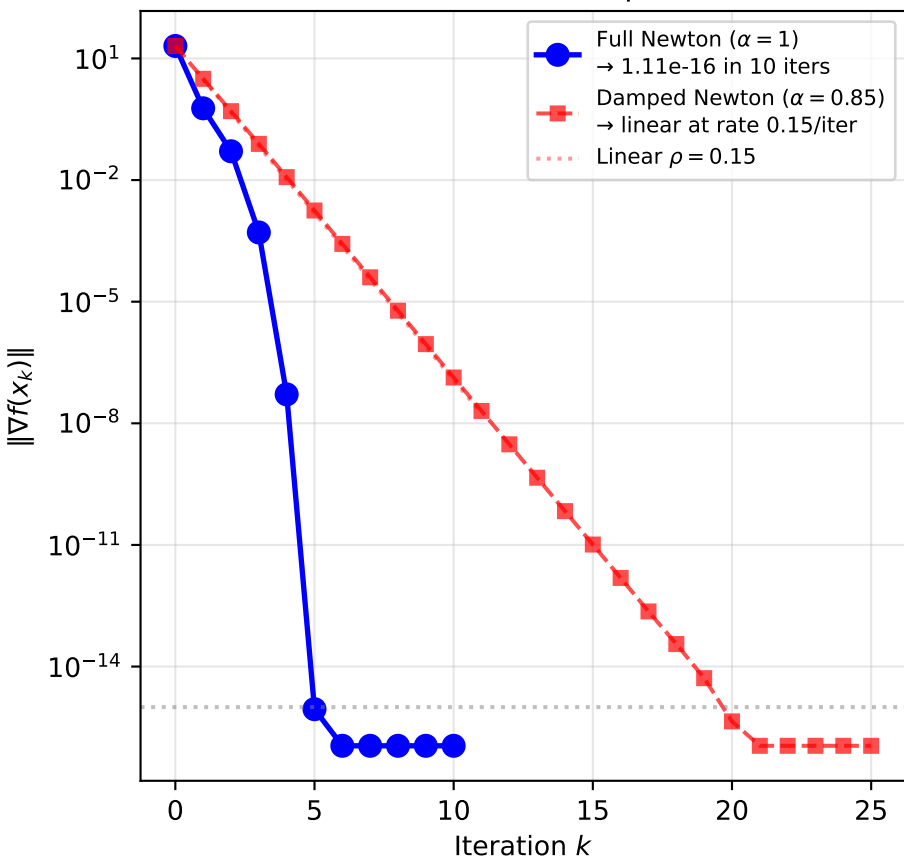


Newton Method Convergence: Quadratic vs Linear (Damped) Rate

Full Newton ($\alpha = 1$):
Quadratic "Digits Doubling"



Full vs Damped Newton
Gradient Norm Comparison



Newton Convergence Data
(Bench~B, green = superlinear phase)

k	$\ \nabla f\ $ (full)	Digits	$\ \nabla f\ $ (damp)
0	2.03e+01	0.0	2.03e+01
1	5.88e-01	0.2	3.14e+00
2	5.13e-02	1.3	5.01e-01
3	5.10e-04	3.3	7.80e-02
4	5.17e-08	7.3	1.18e-02
5	8.88e-16	15.0	1.77e-03
6	1.11e-16	15.7	2.66e-04
7	1.11e-16	15.7	3.99e-05